

Sheet (1)

(1) List the octal and hexadecimal numbers from 16 to 32. Using A, B, and C for the last three digits. list the numbers from 8 to 28 in base 13.

(2) What is the exact number of bytes in a system that contains:

- (a) 32K bytes (b) 64M bytes. (c) 6.4G bytes

(3) Convert the following numbers with the indicated bases to decimal:

- (a) $(4310)_5$ (b) $(198)_{12}$ (c) $(735)_8$ (d) $(525)_{16}$

(4) Express the following numbers in decimal:

- (a) $(10110.0101)_2$ (b) $(16.5)_{16}$ (c) $(26.24)_8$
(d) $(FAFA)_{16}$ (e) $(1010.1010)_2$

(5) Add and multiply the following numbers without converting them to decimal.

- (a) Binary numbers 1011 and 101 (b) Hexadecimal numbers 2E and 34.

(6) Do the following conversion problems:

- (a) Convert decimal 27.315 to binary.
(b) Calculate the binary equivalent of $2/3$ out to eight places. then convert from binary to decimal. How close is the result to $2/3$?
(c) Convert the binary result in (b) into hexadecimal. Then convert the result to decimal. Is the answer the same?

(7) Convert the hexadecimal number 68BE to binary, and then convert it from binary to octal.

(8) Convert the decimal number 431 to binary in two ways:

- (a) Convert directly to binary.
(b) Convert first to hexadecimal and then from hexadecimal to binary.

Which method is faster?

(9) Convert the following binary numbers to hexadecimal and to decimal:

- (a) 1.10010 (b) 110.010

(10) What is the largest binary number that can be expressed with 14 bits? What are the equivalent decimal and hexadecimal number?